Sarthak Verma | EE23B104 | | | My Portfolio

Indian Institute of Technology, Madras

Education

Program	Institution	CGPA/%	Year of Completion
B.Tech in Electrical Engineering	Indian Institute of Technology Madras	8.57/10	2027
Std. XII (CBSE)	Holy Mission Secondary School, Patna	96.8%	2023
Std. X (CBSE)	Loyola High School, Patna	98.2%	2021

Scholastic Achievements

- Secured a position in **top 5** in the IMC Trading simulation, an algorithmic trading competition among 150+ teams.
- Secured a position in **top 30** in Optiver tradeathon with 300+ participants at IIT Madras.
- Achieved a respectable All India Rank in JEE Advanced & JEE Mains 2023 among more than 1.2 million candidates.
- Achieved a repectable rank in KVPY, 2022 conducted by the **IISC, Bangalore** among 50k+ aspirants.
- Recipient of the prestigious **NTSE** Fellowship by the **Govt of India** in the year 2021.
- Ranked in **National Top 1%** in NSEC 2022 and **State Top 1%** in NSEJS 2019, NSEP 2022 and NSEC 2022.
- **State Rank 1**, **Pythagoras Star** - State-level maths contest by **Connectech - UEM** and govt. collaboration.
- Achieved a **National Rank 3** in Montfort International Mathematics Olympiad, 2019.
- Secured an **International Rank of 28** and **Zonal Rank 3** in SOF International Mathematics Olympiad, 2020.
- Secured **Zonal Rank 1** and **International Rank 71** in Silverzone International Informatics Olympiad, 2019.

Course Work & Skills

Stochastic Modelling and theory of Queues	Machine Learning Foundations	Deep Learning
Probability Foundations for Electrical Engineers	Linear Algebra	Multivariable Calculus
Advanced Game theory	Statistics for Electrical Engineers	Numerical Methods with C

Professional Experience

- **Quantitative Research Intern | Goldman Sachs** *May '26 - Jul '26*
  - Engineered calibration pipelines using **Stochastic Volatility Local Volatility (SVLV)** models for **KRW CMS - CMT** exotics, optimizing volatility surface fitting and front-office pricing accuracy.
  - Architected a market-benchmarking engine to validate exotic rate pricing against **Totem consensus data**, directly enhancing desk quoting precision and mark-to-market accuracy.
  - Developed a real-time risk & diagnostics engine to automate PnL explain, Greeks/risk attribution, and scenario decomposition, driving decision support for exotic rate traders.
- **CYBATHLON Competition | Zurich, Switzerland** *October 2024*
  - Represented IIT Madras and the **R2D2**** **Lab** at an international competition focused on assistive technologies.
  - Achieved a **global rank of 8**, marking the first Indian student-led team to compete in this domain.
  - Designed real-time control systems and embedded solutions for assistive devices focused on above-knee amputees.

Research & Publications

- **Metaoptics - GUI-Driven Metaoptical Design Tool in Python** *Jan '25 - Present*
  - **Paper Submitted:** Co-authored research paper submitted to **Journal of Optics** (**Springer Nature**, official journal of the Optical Society of India) under the guidance of **Prof. Shanti Bhattacharya**.
  - Developed Python software contributing to fast and efficient generation of GDS files for Metamaterial fabrication.
  - Supported hexagonal lattices and **dual-wavelength phase profiling** via variable-radius nanopillars.

Projects

- **Transformer-Based Stock Return Forecasting** *Jan '26 - Mar '26*
  - Developed a **Transformer forecasting model** in PyTorch to capture long-range dependencies in time series.
  - Engineered **price, volume, momentum, and volatility features** with sliding-window sampling, positional encoding, multi-head self-attention, and multi-horizon return forecasting using walk-forward validation.
- **Trading Strategy Arena - Quantitative Backtesting & Signal Engine** *May '25 - Jul '25*
  - Built an **event-driven backtesting engine** in Python to benchmark multi-agent strategies via Tkinter.
  - Modeled dynamic market frictions (slippage, TCA, execution latency) to eliminate systematic lookahead bias.
  - Evaluated systematic StatArb and multi-factor ML signals, tracking Sharpe ratio, Alpha/Beta, and rolling VaR.
- **Simulation and Optimization in Python (Course Project)** *Jul '24 - Nov '24*
  - Leveraged Python to build a **SPICE circuit simulator** and optimized topologies via **Simulated Annealing**.
  - Profiled bottlenecks and implemented **Cython** C-extensions to accelerate performance-critical execution routines.
- **Numerical Methods with C (Course Project)** *Jul '23 - Nov '23*
  - Implemented root-finding, Lagrangian/spline interpolation, Gaussian Elimination, and LU Decomposition solvers.
  - Solved non-linear **Landau-Lifshitz-Gilbert ODEs** via RK, Heun, and Euler-Cauchy for magnetic switching.
  - Generated pseudo-random sequences, computed Hamming distances, and validated randomness via NIST suite.

Leadership & Entrepreneurship

- **LeetCore - Founder** *May '26 - Present*
  - Engineered and scaled an online assessment platform to **2,000+ active users** with proctored contest engines.
  - Formed a partnership with **GlobalFoundries** to integrate industry-aligned technical evaluation modules.
  - Deployed full-stack infrastructure at **leetcore.co.in** featuring real-time analytics and mock exam suites.
- **Head EE UG, Placement and Internship Cell** *May '26 – present*
  - Leading a team of 14 for the upcoming placement and internship for the EE department at IITM.
  - Coordinated with 20+ Companies and 40+ Profiles as Primary/Secondary POC during the previous season.
  - Developed and implemented placement policies and guidelines for students and recruiters at IIT Madras.

** Rehabilitation Research and Device Development